

CPI-Based Household Budget Forecasting Using Machine Learning and Time Series Models

A Project Report

Abstract

Inflation directly affects the purchasing power of consumers by increasing the prices of goods and services over time. Traditional budgeting methods assume that monthly expenses remain constant, often leading to unrealistic financial planning during periods of inflation. This project proposes an intelligent budget forecasting system that predicts future Consumer Price Index (CPI) values and estimates how household budgets are expected to change over the next six months.

Three forecasting models—Ridge Regression, SARIMA, and SARIMAX—were developed independently to predict CPI for six major expenditure categories:

- Food
- Housing
- Transportation
- Medical
- Education
- Energy

The predicted CPI values are integrated into a Streamlit-based web application where users enter their current monthly expenses. The system adjusts each expense category according to the predicted inflation, allowing users to visualize their expected future budget.

Unlike conventional forecasting projects that stop after generating predictions, this project converts inflation forecasts into practical financial recommendations, making the work directly applicable to household financial planning.

Chapter 1: Introduction

Background

Inflation is one of the most important economic indicators influencing the cost of living. Even moderate inflation gradually reduces purchasing power, causing households to spend more money for the same goods and services.

Most individuals prepare monthly budgets assuming stable prices. However, inflation changes expenditure patterns continuously.

Therefore, predicting future CPI enables households to estimate:

- How much more they may need to spend
- Which expense categories will experience the largest increase
- Overall increase in monthly expenditure

This project bridges economic forecasting with personal finance by combining predictive analytics with an interactive budgeting tool.

Problem Statement

Households generally prepare budgets using current prices.

They lack a mechanism to estimate:

- Future food costs
- Future housing expenses
- Medical inflation
- Transportation inflation

Consequently, many households underestimate future spending requirements.

The project aims to solve this problem by forecasting CPI and translating inflation forecasts into personalized future budgets.

Objectives

- Collect historical CPI data
- Forecast category-wise CPI for six months
- Compare three forecasting approaches
- Build an interactive budgeting application
- Provide inflation-adjusted budget estimates
- Visualize projected expenditure trends
- Help users understand the financial impact of inflation

Scope

The project focuses on six expenditure categories:

- Food
- Housing
- Transportation
- Education
- Medical
- Energy

Forecast horizon: 6 Months

Forecasting Models:

- Ridge Regression
- SARIMA
- SARIMAX

Deployment: Streamlit Web Application

Chapter 2: Literature Review

Inflation forecasting has traditionally relied upon statistical time-series techniques. Common methods include:

ARIMA

Captures autoregressive relationships and moving averages.

Advantages:

- Well-established
- Effective for stationary data

Limitations:

- Limited ability to incorporate external variables.

SARIMA

Extends ARIMA by modelling seasonality.

Advantages:

- Handles yearly seasonal CPI patterns
- Suitable for monthly economic data

Limitations:

- Uses only historical CPI values.

SARIMAX

Further extends SARIMA by allowing external explanatory variables.

Advantages:

- Incorporates additional information
- Often produces more robust forecasts

Ridge Regression

Ridge Regression is a regularized linear regression model that reduces overfitting by penalizing large coefficients.

Advantages:

- Handles multicollinearity
- Stable predictions
- Fast training
- Easy interpretation

Chapter 3: Dataset

The dataset consists of historical monthly CPI values.

Each row represents one month.

The dataset contains six CPI categories:

Category	Description
Food	Food inflation
Housing	Housing CPI
Transportation	Transport CPI
Medical	Healthcare CPI
Education	Education CPI
Energy	Energy prices

The Date column serves as the time index.

Chapter 4: Methodology

Step 1: Data Collection

Historical CPI values were collected and organized into monthly observations.

The data was cleaned by:

- Converting dates
- Ensuring monthly frequency
- Removing inconsistencies
- Preparing category-wise datasets

Step 2: Ridge Regression Forecasting

For every CPI category, the following features were generated:

- Lag-1 CPI
- Lag-12 CPI
- Previous monthly difference
- Month sine encoding
- Month cosine encoding

These features allow the regression model to learn:

- Trend
- Seasonality
- Recent movement

RidgeCV automatically selected the optimal regularization parameter through cross-validation. After training, six-month forecasts were generated.

Step 3: SARIMA Forecasting

Each CPI category was modelled independently. SARIMA captures:

- Autoregression
- Differencing
- Moving averages
- Yearly seasonality

A separate model was fitted for each expenditure category before forecasting six future months.

Step 4: SARIMAX Forecasting

SARIMAX follows the same framework as SARIMA while incorporating additional explanatory variables when available. A parameter search was performed to identify the best seasonal and non-seasonal orders before generating the six-month forecasts.

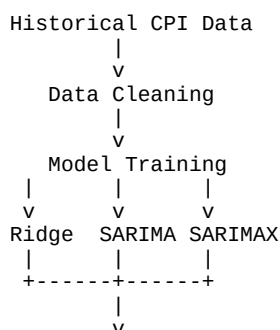
Step 5: Budget Projection

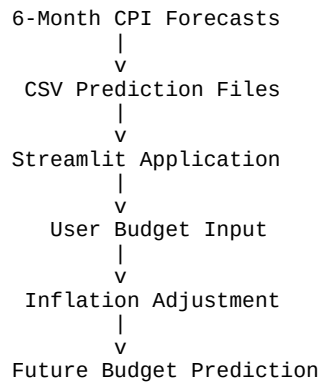
The forecasting models produce predicted CPI values rather than budgets. To convert CPI into budget estimates, the application scales the user's current spending by the ratio of forecast CPI to the first forecast month:

$$\text{Projected Budget} = \text{Current Budget} \times (\text{Forecast CPI}_t / \text{Forecast CPI}_{\text{base}})$$

This maintains the user's spending proportions while adjusting them for projected inflation.

Chapter 5: System Architecture





Chapter 6: Streamlit Application

The Streamlit application serves as the user interface.

Users enter their monthly expenses for each expenditure category.

The application then:

- Loads CPI forecasts
- Computes projected future budgets
- Compares results across all three models
- Displays tables and visualizations
- Ranks categories by expected CPI increase

The interface contains separate tabs for SARIMA, SARIMAX and Ridge Regression, loads forecast files, projects category budgets using CPI ratios, and presents charts including projected budgets, normalized comparisons, ranked CPI changes, and detailed forecast tables.

Visualizations

1. Projected Budget per Category

Shows how each spending category changes over six months.

2. Indexed Comparison Chart

Normalizes all categories to compare growth rates regardless of absolute budget.

3. CPI Change Ranking

Ranks expenditure categories according to expected inflation.

4. Budget Projection Table

Shows projected monthly expenses numerically.

5. Underlying CPI Forecast

Displays the predicted CPI values used for budget calculations.

Chapter 7: Results

The developed models successfully generated six-month CPI forecasts for all expenditure categories. The budgeting application translated these forecasts into projected household expenses.

The system demonstrates that:

- Inflation affects spending categories differently
- Future budgets increase gradually over time
- Users can compare multiple forecasting methods
- Inflation forecasts become actionable financial information

Applications

This project can be used by:

- Individual households
- Financial planners
- Students
- Budgeting applications
- Personal finance platforms
- Government awareness initiatives
- Banking and fintech companies

Advantages

- Combines statistical forecasting with machine learning.
- Uses three independent forecasting techniques.
- Interactive and user-friendly interface.
- Category-wise expenditure analysis.
- Supports proactive financial planning.
- Easily extendable to additional expenditure categories.

Limitations

- Forecast accuracy depends on historical CPI behavior.
- Unexpected economic shocks (pandemics, wars, policy changes) are not captured.
- Budget estimates assume that spending patterns remain constant apart from inflation.
- Only a six-month forecasting horizon is considered.

Future Work

- LSTM or Transformer-based forecasting models.
- Automatic monthly data updates via APIs.
- Confidence intervals for CPI forecasts.
- Personalized spending pattern learning.
- Income forecasting alongside expenses.
- Inflation scenario analysis (optimistic, expected, pessimistic).
- Mobile application deployment.
- Integration with bank transaction data.

Conclusion

This project presents a complete inflation-aware budgeting framework that combines economic forecasting with practical financial planning. By forecasting category-wise Consumer Price Index values using Ridge Regression, SARIMA, and SARIMAX, and integrating these forecasts into an interactive Streamlit application, the system transforms abstract inflation predictions into personalized budget projections.

The project demonstrates that predictive analytics can support everyday financial decision-making by helping users anticipate future expenses, compare forecasting models, and identify categories most affected by inflation. Beyond being a forecasting exercise, it provides a practical decision-support tool that can be extended for use in personal finance, fintech, and household budgeting applications.